

# SEE6213: WASTEWATER ENGINEERING AND WATER QUALITY ASSESSMENT

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## Effective Term

Semester A 2025/26

## Part I Course Overview

### Course Title

Wastewater Engineering and Water Quality Assessment

### Subject Code

SEE - School of Energy and Environment

### Course Number

6213

### Academic Unit

School of Energy and Environment (E2)

### College/School

School of Energy and Environment (E2)

### Course Duration

One Semester

### Credit Units

3

### Level

P5, P6 - Postgraduate Degree

### Medium of Instruction

English

### Medium of Assessment

English

### Prerequisites

Nil

### Precursors

Nil

### Equivalent Courses

SEE8221 Wastewater Engineering and Water Quality Assessment

### Exclusive Courses

Nil

## Part II Course Details

### Abstract

The course aims to provide students with the fundamental knowledge on wastewater engineering processes as well as the analytical techniques involved in assessing water quality. State-of-the-art processes in wastewater treatment will also be covered in the course.

### Course Intended Learning Outcomes (CILOs)

| CILOs |                                                                                                                                                  | Weighting (if DEC-A1 app.) |   | DEC-A2 | DEC-A3 |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|---|--------|--------|
| 1     | Identify and classify the different sources of wastewater and their requirement for treatment depending on their discharge or final utilisation. | 10                         | x |        |        |
| 2     | Describe and perform various analysis of water and wastewater quality assessment.                                                                | 20                         |   | x      |        |
| 3     | Design the various physical and chemical unit operations for wastewater treatment.                                                               | 20                         |   | x      |        |
| 4     | Design the various biological unit operations for wastewater treatment.                                                                          | 20                         |   | x      |        |
| 5     | Describe the principles of various advanced treatment, concepts of water recycling and desalination.                                             | 30                         |   |        | x      |

#### A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

#### A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

#### A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

### Learning and Teaching Activities (LTAs)

| LTAs |          | Brief Description                                                                                                                                                                                       | CILO No.   | Hours/week (if applicable) |
|------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------------------------|
| 1    | Lectures | The lectures will cover basic theories and concepts of wastewater treatment and control methods, techniques and operations of waste treatment plants, and management systems commonly use in Hong Kong. | 1, 2, 3, 4 | 2 hrs/week                 |
| 2    | Tutorial | Open discussions in tutorial sessions will be given to students on engineering calculation procedure and formulation techniques.                                                                        | 2, 3, 4    | 1 hr/week                  |

|   |               |                                                                                                        |   |  |
|---|---------------|--------------------------------------------------------------------------------------------------------|---|--|
| 3 | Laboratory    | Analysis water quality and understand its impact on environment                                        | 2 |  |
| 4 | Field visit   | Recognize the contemporary technology at national levels in addressing environment problems and issues | 4 |  |
| 5 | Mini projects | Develop innovative and creative solutions to wastewater treatment through teamwork and projects        | 5 |  |

**Assessment Tasks / Activities (ATs)**

|   | ATs         | CILO No.   | Weighting (%) | Remarks ("- for nil entry) | Allow Use of GenAI? |
|---|-------------|------------|---------------|----------------------------|---------------------|
| 1 | Assignments | 2, 3, 4    | 20            | -                          | Yes                 |
| 2 | Project     | 5          | 30            | -                          | Yes                 |
| 3 | Quiz        | 1, 2, 3, 4 | 20            | -                          | No                  |

**Continuous Assessment (%)**

70

**Examination (%)**

30

**Examination Duration (Hours)**

2

**Minimum Continuous Assessment Passing Requirement (%)**

30

**Minimum Examination Passing Requirement (%)**

30

**Additional Information for ATs**

To pass a course, a student must do ALL of the following:

- 1) obtain at least 30% of the total marks allocated towards coursework (combination of assignments, pop quizzes, term paper, lab reports and/ or quiz, if applicable);
- 2) obtain at least 30% of the total marks allocated towards final examination (if applicable); and
- 3) meet the criteria listed in the section on Assessment Rubrics.

**Assessment Rubrics (AR)****Assessment Task**

Assignments (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

**Criterion**

Graded assignment in each topic with both calculations and structured problem solving

**Excellent**

(A+, A, A-) The student completes all assessment tasks/activities and the work demonstrates excellent understanding of the scientific principles and the working mechanisms

**Good**

(B+, B, B-) The student completes all assessment tasks/activities and can describe and explain the scientific principles.

**Fair**

(C+, C, C-) The student completes all assessment tasks/activities and can describe and explain some scientific principles.

**Marginal**

(D) The student completes all assessment tasks/activities but can only briefly describe some scientific principles.

**Failure**

(F) The student fails to complete all assessment tasks/activities

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**Assessment Task**

Test (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

**Criterion**

Comprehensive paper examination with both calculations and structured problem solving in evaluating student' s learning abilities

**Excellent**

(A+, A, A-) He/she can thoroughly identify and explain how the principles are applied to wastewater engineering.

**Good**

(B+, B, B-) He/she provides a detailed evaluation of how the principles are applied to wastewater engineering.

**Fair**

(C+, C, C-) He/she provides simple but accurate evaluations of how the principles are applied to wastewater engineering.

**Marginal**

(D) Only some of the analysis is appropriate to show how the principles are applied to wastewater engineering.

**Failure**

(F) He/she fails to identify and explain how the principles are applied to wastewater engineering.

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**Assessment Task**

Mini Project (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

**Criterion**

Analyse and provide innovative engineering solution in wastewater treatment

**Excellent**

(A+, A, A-) He/she is able to communicate ideas effectively and persuasively via written texts and/or oral presentation.

**Good**

(B+, B, B-) He/she is able to communicate ideas effectively via written texts and/or oral presentation.

**Fair**

(C+, C, C-) He/she can communicate ideas clearly in written texts and/or in oral presentations.

**Marginal**

(D) He/she can communicate simple ideas in writing and/or in oral presentations.

### **Failure**

(F) He/she is weak in communicating ideas and/or the student's work shows evidence of plagiarism.

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### **Assessment Task**

Assignments (for students admitted from Semester A 2022/23 to Summer Term 2024)

### **Criterion**

Graded assignment in each topic with both calculations and structured problem solving

### **Excellent**

(A+, A, A-) The student completes all assessment tasks/activities and the work demonstrates excellent understanding of the scientific principles and the working mechanisms

### **Good**

(B+, B) The student completes all assessment tasks/activities and can describe and explain the scientific principles.

### **Marginal**

(B-, C+, C) The student completes all assessment tasks/activities but can only briefly describe some scientific principles.

### **Failure**

(F) The student fails to complete all assessment tasks/activities

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### **Assessment Task**

Test (for students admitted from Semester A 2022/23 to Summer Term 2024)

### **Criterion**

Comprehensive paper examination with both calculations and structured problem solving in evaluating student's learning abilities

### **Excellent**

(A+, A, A-) He/she can thoroughly identify and explain how the principles are applied to wastewater engineering.

### **Good**

(B+, B) He/she provides a detailed evaluation of how the principles are applied to wastewater engineering.

### **Marginal**

(B-, C+, C) Only some of the analysis is appropriate to show how the principles are applied to wastewater engineering.

### **Failure**

(F) He/she fails to identify and explain how the principles are applied to wastewater engineering.

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### **Assessment Task**

Mini Project (for students admitted from Semester A 2022/23 to Summer Term 2024)

### **Criterion**

Analyse and provide innovative engineering solution in wastewater treatment

### **Excellent**

(A+, A, A-) He/she is able to communicate ideas effectively and persuasively via written texts and/or oral presentation.

**Good**

(B+, B) He/she is able to communicate ideas effectively via written texts and/or oral presentation.

**Marginal**

(B-, C+, C) He/she can communicate simple ideas in writing and/or in oral presentations.

**Failure**

(F) He/she is weak in communicating ideas and/or the student's work shows evidence of plagiarism.

## Part III Other Information

**Keyword Syllabus**

- Composition and classification of wastewater
- Analytical techniques in water quality assessment
- Physical unit operations - sedimentation, flocculation, flotation
- Chemical unit operations - aeration, pH, chlorination, ion exchange
- Biological treatment – aerobic/anaerobic treatment, activated sludge, trickling filter
- Membrane bioreactor (MBR)
- Advanced oxidation processes – ozone, UV, Fenton, photo-Fenton, photocatalysis
- Water recycling and desalination

**Reading List****Compulsory Readings**

| Title |                                                                                                                                                 |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Metcalf and Eddy /Aecom, Wastewater Engineering: Treatment and Resource Recovery 5th Edition, McGraw-Hill International Edition, 2014.          |
| 2     | MIHELICIC, J.R. and ZIMMERMAN, J.J. (2009) Environmental Engineering: Fundamentals, Sustainability, Design. New Jersey: John Wiley & Sons, Ltd. |

**Additional Readings**

| Title |                                                                                                                     |
|-------|---------------------------------------------------------------------------------------------------------------------|
| 1     | MACKENZIE , D. (2010) Water and Wastewater Engineering. New York: McGraw-Hill, Ltd.                                 |
| 2     | METCALF & EDDY: AECOM, Inc. (2007) Water Reuse: Issues, Technologies, and Applications, New York: McGraw-Hill, Ltd. |
| 3     | IZRAIL, S., TUROVSKIY, P. and MATHAI, K. (2006) Wastewater Sludge Processing. New Jersey: John Wiley & Sons, Ltd.   |
| 4     | RUSSELL, D.L. (2006) Practical Wastewater Treatment. New Jersey: John Wiley & Sons, Ltd.                            |